

Programming Assignment 01

MD Shantanu Rahman

Texas State University

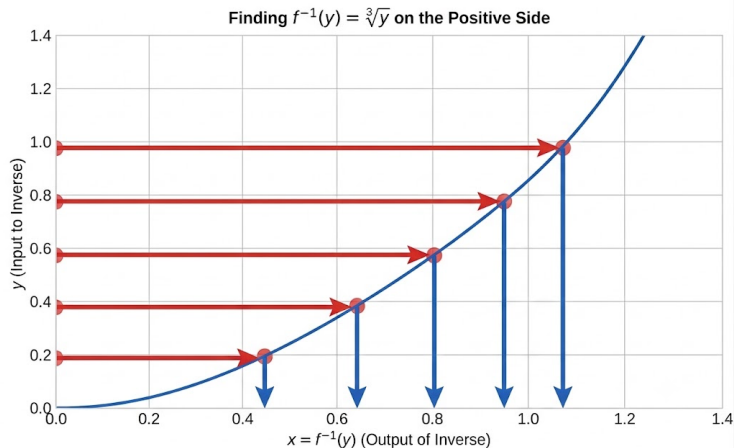
September 3, 2026

Programming Assignment

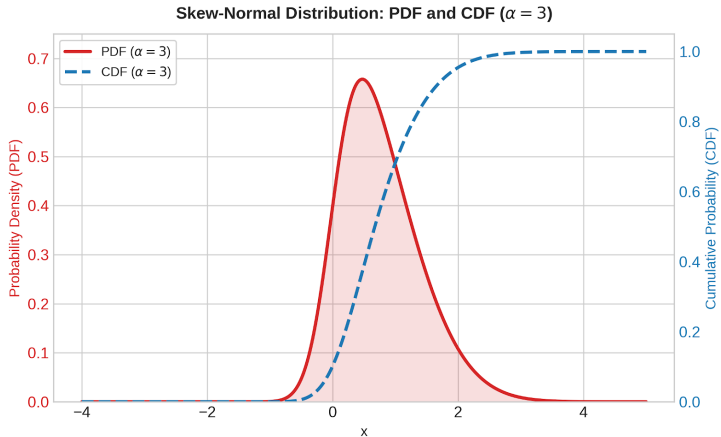
<https://canvas.txstate.edu/courses/2736540/assignments/42813946>

Inverse Transform Method

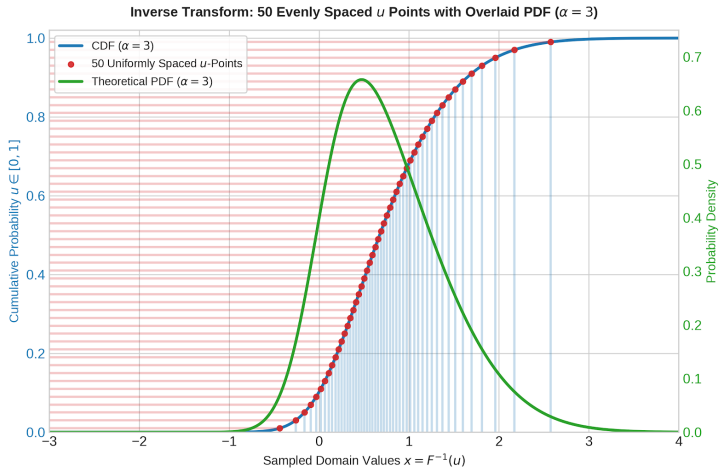
Consider $f(x) = x^3$, How would we graphically figure out $f^{-1}(x)$?



Inverse Transform Method



Inverse Transform Method



Inverse Transform Method

The Cumulative Distribution Function of an exponentially distributed random variable x ,

$$F(x) = 1 - \exp(-\lambda x)$$

Inverse Transform Method

$$F(F^{-1}(u)) = u$$

$$1 - \exp(-\lambda F^{-1}(u)) = u$$

$$F^{-1}(u) = -\frac{1}{\lambda} \ln(1 - u)$$

$$T_s = \frac{1}{\lambda}$$

Inverse Transform Method

- ▶ Generate a number x using an uniform distribution
- ▶ use the F^{-1} function to calculate $y = F^{-1}(x)$
- ▶ $y \in \text{Exponential}(\lambda)$

Questions?

Resources

- [1] https://en.wikipedia.org/wiki/Probability_integral_transform
- [2] <https://youtu.be/TzKANDzAXnQ>

Thanks